NutriLens: An AI-Based Nutrition and Dietary Assessment App

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Chapter 1

Introduction

The product described in this document is NutriLens, a mobile AI nutrition assistant designed to help users make healthier and safer food choices. NutriLens integrates OCR, NLP, and machine learning to scan packaged food products, check for allergens and halal compliance, calculate personalized Nutri-Scores, predict health risks, provide dietary guidance via a chatbot, and present analytics through an interactive dashboard.

1.1 Problem Statement

Modern consumers face challenges in making safe and informed food choices. Many packaged food items contain allergens, non-halal ingredients, or additives that may be harmful, yet users have no easy way to check these details. Additionally, existing nutrition tracking apps are often generic, fail to consider personal health profiles, and do not provide culturally relevant guidance. People with dietary restrictions, chronic illnesses, or specific health goals struggle to interpret nutrition labels accurately. Manual food logging is tedious, prone to error, and lacks predictive insights for health risks such as diabetes, hypertension, or obesity. NutriLens addresses these issues by automating label scanning, performing ingredient compliance checks, and generating personalized nutrition insights. By leveraging AI and machine learning, the system predicts health risks based on real-time consumption patterns and provides actionable advice through a chatbot and visual dashboards. The app aims to reduce diet-related health risks, promote safer food choices, and empower users to maintain healthier eating habits.

1.2 Scope

NutriLens is a mobile application designed for individuals seeking personalized nutrition guidance and food safety alerts. Users can scan packaged food labels to automatically extract ingredients and nutritional information using OCR. The system evaluates ingredients for allergens, harmful additives, and halal compliance using NLP techniques. Based on the extracted nutrition data, NutriLens calculates personalized Nutri-Scores using machine learning models that consider user health profiles, BMI, age, and dietary restrictions. The app logs food intake automatically or manually and continuously monitors the 24-hour consumption pattern to predict health risks like obesity or hypertension. Users receive alerts and guidance via an AI Nutrition Coach, which provides chat-based advice, portion recommendations, and context-aware tips. The interactive dashboard visualizes nutrient trends, progress toward health goals, and risk alerts. The system focuses on packaged food products and does not currently handle restaurant meals or fresh produce. Overall, NutriLens combines AI-driven analysis with an intuitive interface to enhance health awareness, support informed dietary decisions, and accommodate cultural and religious food requirements.

1.3 Modules

The NutriLens system is divided into five main modules, each handling a specific function of the application.

1.3.1 Module 1: Product Scanning and Data Extraction

This module enables users to scan packaged food items and extract nutrition and ingredient information using OCR technology. It ensures accurate text extraction and handles errors from poor-quality scans.

1. OCR-based text extraction from product labels.
2. Automatic parsing of nutrition facts and ingredients.
3. Handling errors for incomplete or low-quality scans.

1.3.2 Module 2: Ingredient Analysis and Compliance

This module processes extracted ingredient data to ensure dietary safety, allergen detection, and halal compliance. NLP techniques are used for efficient and accurate analysis.

1. Detect allergens according to user-specific dietary restrictions.
2. Identify harmful additives and E-codes.
3. Verify halal compliance for ingredients.

1.3.3 Module 3: Personalized Nutri-Score and Health Prediction

This module calculates personalized Nutri-Scores and predicts health risks using machine learning models trained on the NHANES dataset. It integrates user health profiles for tailored insights.

1. Calculate personalized Nutri-Scores based on age, BMI, and health profile.
2. Predict health risks like diabetes, hypertension, and obesity from consumption trends.
3. Provide visual indicators for unhealthy eating patterns.

1.3.4 Module 4: AI Nutrition Coach

This module offers interactive guidance to help users maintain healthier eating habits. It provides personalized advice and portion control recommendations via a chatbot interface.

1. Chat-based nutrition advice.
2. Personalized portion size and calorie intake recommendations.
3. Context-aware suggestions for balanced diet planning.

1.3.5 Module 5: Dashboard and Analytics

This module presents the user's dietary data, progress trends, and risk alerts in a visual format. It allows users to monitor their nutrition intake over time and make informed decisions.

1. Graphical dashboard for nutrient trends and calorie consumption.
2. Track progress against personal health goals.
3. Display alerts for detected health risks or unhealthy consumption patterns.

1.4 User Classes and Characteristics

User class	Description
End User	An End User is an individual who wants to monitor their nutrition, check food for allergens or halal compliance, and receive personalized dietary guidance. Users will scan packaged food products or manually log meals. They may use the app daily to track consumption and monitor health risks. The app is designed for people of all ages with varying health conditions and dietary restrictions.
Nutrition Coach	The AI Nutrition Coach is an intelligent chatbot agent that interacts with End Users through the chat interface. It provides dietary guidance, portion recommendations, and tips for healthier eating.
System Administrator	A System Administrator manages the backend infrastructure, ensuring the database, OCR/NLP pipelines, and ML models function correctly. Administrators may update ingredient dictionaries, add new compliance rules, or maintain the ML model and data. Typical interactions are backend-focused and not directly user-facing.
Healthcare Professional	Optional user class for future expansion. Healthcare Professionals may monitor patient dashboards, review risk alerts, and give medical advice based on aggregated nutrition and health data. They require access to secure analytics and reporting tools.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-Case Representation

For NutriLens, a Use Case approach is most suitable because it is an interactive end-user application. The system involves scanning products, analyzing ingredients, generating Nutri-Scores, logging meals, and interacting with an AI chatbot. Each functionality is triggered by user actions and the system responds with processed results. Therefore, use case diagrams and detailed use cases are the appropriate requirement gathering technique.

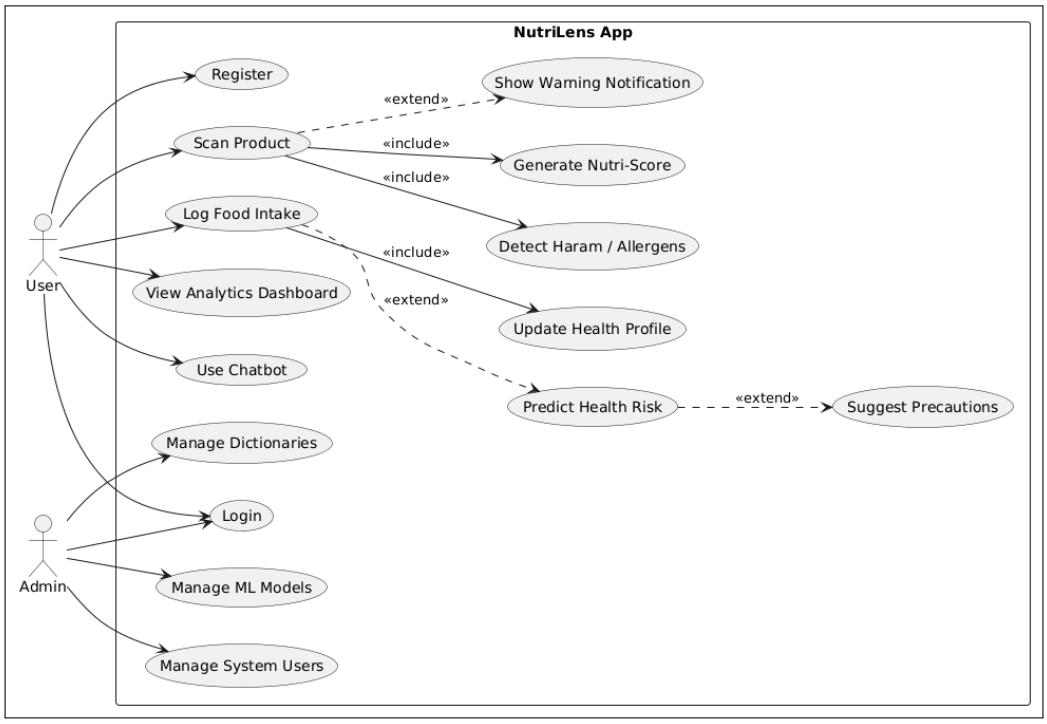

Figure 2.1: Use Case Diagram of NutriLens

2.2 Functional Requirements

This section describes the functional requirements of the system expressed in natural language style. Requirements are organized by system modules.

2.2.1 Module 1: Product Scanning and Data Extraction

Following are the requirements for module 1:

1. FR1: The system shall allow users to scan packaged food labels using the mobile camera.
2. FR2: The system shall extract ingredient and nutrition text from the scanned image using OCR.
3. FR3: The system shall automatically parse the extracted text into structured ingredient lists and nutrition values.
4. FR4: The system shall handle errors from poor-quality or incomplete scans and prompt the user to rescan.

2.2.2 Module 2: Ingredient Analysis and Compliance

Following are the requirements for module 2:

1. FR1: The system shall analyze the scanned ingredient list to detect allergens based on user settings.
2. FR2: The system shall identify harmful additives, E-codes, and non-halal ingredients in the list.
3. FR3: The system shall generate a report indicating whether the scanned product is safe, contains allergens, or is non-halal.
4. FR4: The system shall allow users to view alerts or warnings for detected issues.

2.2.3 Module 3: Personalized Nutri-Score and Health Prediction

Following are the requirements for module 3:

1. FR1: The system shall calculate a personalized Nutri-Score based on the user's age, BMI, health profile, and scanned nutrition data.

2. FR2: The system shall predict health risks like hypertension, or diabetes based on logged food intake and trends.
3. FR3: The system shall display visual indicators for unhealthy consumption patterns.
4. FR4: The system shall save Nutri-Score results and risk flags in the user's profile for dashboard analytics.

2.2.4 Module 4: AI Nutrition Coach

Following are the requirements for module 4:

1. FR1: The system shall provide a chat interface for the AI Nutrition Coach.
2. FR2: The system shall respond to user queries with context-aware dietary advice.
3. FR3: The system shall provide portion recommendations and tips for healthier eating.
4. FR4: The system shall maintain session context for ongoing chat interactions.

2.2.5 Module 5: Dashboard and Analytics

Following are the requirements for module 5:

1. FR1: The system shall display a graphical dashboard showing nutrient intake trends and calorie consumption over time.
2. FR2: The system shall track progress toward user-defined health and nutrition goals.
3. FR3: The system shall display alerts for health risks or unhealthy consumption patterns.
4. FR4: The system shall allow users to filter and view historical food logs and Nutri-Score trends.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements. The requirements are specific, measurable, and verifiable.

2.3.1 Reliability

1. The system shall have a mean time between failures (MTBF) of at least 30 days for the mobile application and backend services.
2. The system shall implement error detection for OCR and NLP processing to prevent incorrect classification of ingredients.
3. The system shall provide fallback mechanisms in case of ML model or database failures, including cached results and user notifications.

2.3.2 Usability

1. The system shall allow a user to scan a product and receive ingredient analysis and Nutri-Score results within 5 seconds.
2. The system shall provide a simple and intuitive chat interface for interacting with the AI Nutrition Coach.
3. The system shall allow users to view historical nutrition data and trends with a single interaction from the dashboard.
4. The system shall provide clear visual alerts for allergens, non-halal ingredients, or health risk flags.

2.3.3 Performance

1. The OCR extraction shall process scanned images and return structured text in under 3 seconds for standard label images.
2. The dashboard shall render graphs and trend analytics within 2 seconds for up to 30 days of user data.

2.3.4 Security

1. The system shall protect user data (personal health information, food logs) using encrypted storage on the device and backend.
2. The system shall require secure authentication for access to user profiles and sensitive data.
3. The system shall prevent unauthorized access to ML model results and health risk predictions.

4. The system shall maintain privacy by ensuring user data is not shared without consent.

2.4 Domain Model

The following diagram represents the domain model of the NutriLens project, showing the key entities, their attributes, and relationships within the system.

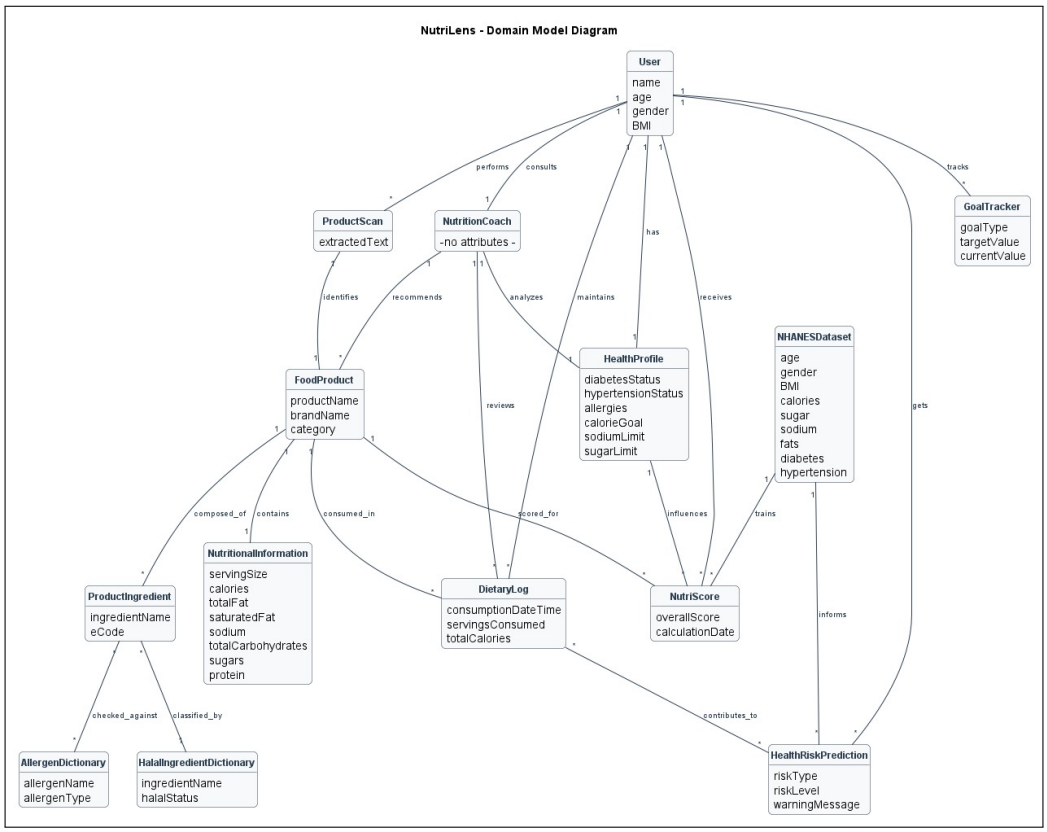

Figure 2.2: Domain Model Diagram of NutriLens

Chapter 3

System Overview

3.1 Architectural Design

The architecture of NutriLens follows a **multi-tier client-server architecture** designed for scalability, maintainability, and modularity. It separates the system into distinct manageable layers, each responsible for specific operations. This architectural design ensures flexibility for future expansions such as barcode scanning, meal planning, community forums, or wearable device integration.

The system architecture consists of the following three layers:

- **Presentation Layer (Client/Mobile Application):** This includes the mobile application interface developed using React Native. It handles interactions with the user such as scanning food products, logging meals, viewing analytics, and chatting with the AI Nutrition Coach. Communication with the backend occurs through RESTful API endpoints.
- **Application Layer (Business Logic and Services):** This layer processes system operations including OCR-based ingredient extraction, halal/allergen detection, calorie calculations, Nutri-Score generation, user authentication, and machine learning predictions. It is implemented using a combination of Node.js microservices and Python FastAPI services.
- **Data Layer (Database and Storage):** MongoDB is used for user data storage, food logs, and chat history.

To ensure good object-oriented design, the system incorporates **GRASP (General Responsibility Assignment Software Patterns)** principles such as:

- **Controller:** Each user request is handled through a centralized controller in the application layer.

- **Creator:** Objects responsible for creating related data are assigned creation responsibilities.
- **Low Coupling and High Cohesion:** Each module performs a focused task and minimizes dependency on other modules.

The following diagrams illustrate the structural organization of the system architecture:

3.1.1 Component Diagram

The component diagram represents the main components of the system and their dependencies including mobile UI, OCR service, AI service, database, and chatbot module.

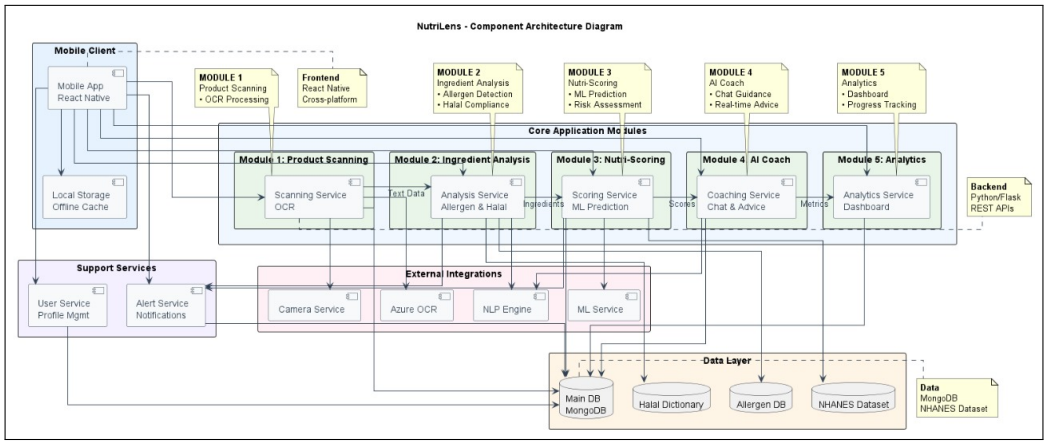

Figure 3.1: Component Diagram of NutriLens

3.1.2 Deployment Diagram

The deployment diagram illustrates the physical architecture of the system including client devices, backend servers, cloud services, and databases.

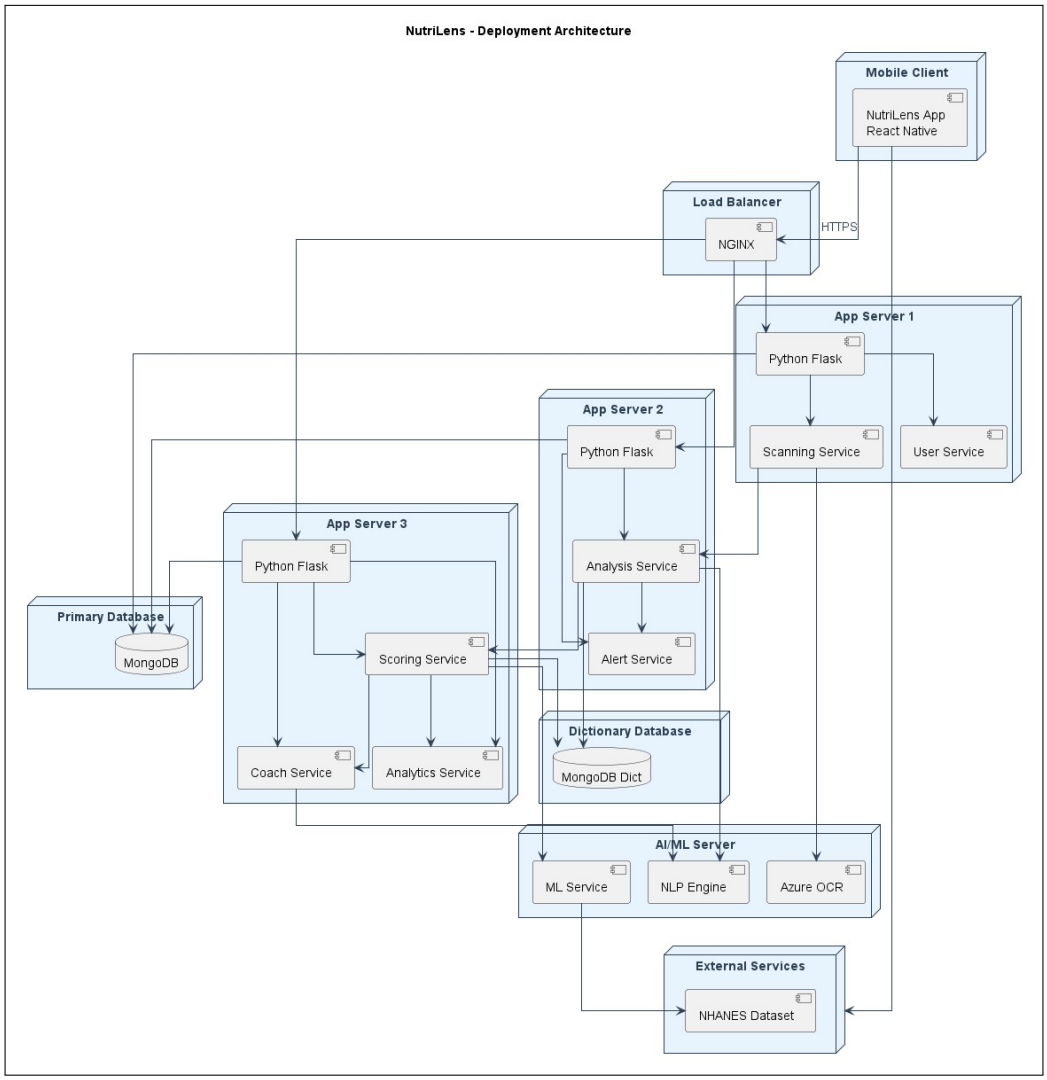

Figure 3.2: Deployment Diagram of NutriLens

3.1.3 Package Diagram

The package diagram shows how the project is organized into logical groups such as UI, Services, Controllers, Models, and Utilities.

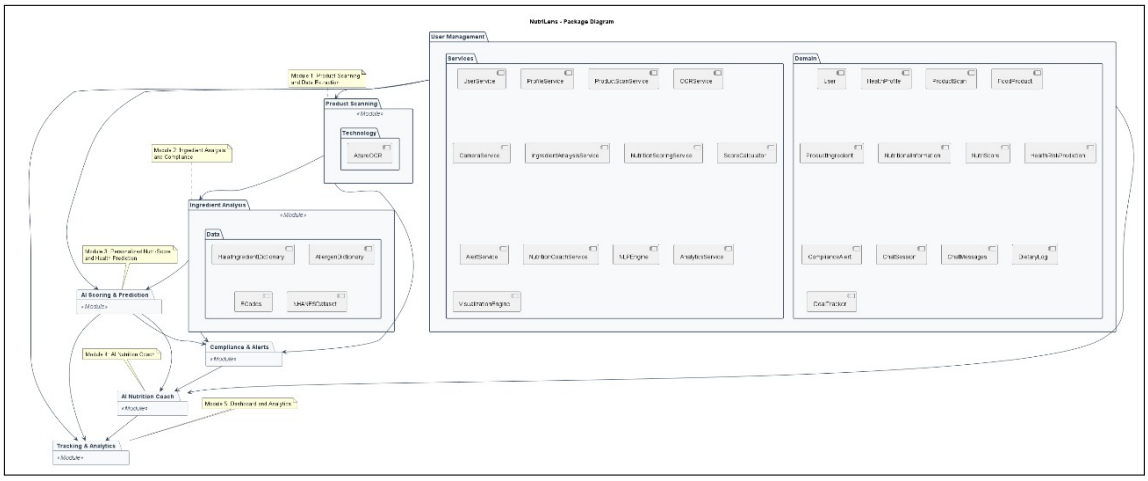

Figure 3.3: Package Diagram of NutriLens

3.2 Design Models

Design models provide structured representations of system behavior and architecture. The models below follow an object-oriented development approach, showing both dynamic and static perspectives of NutriLens.

3.2.1 Activity Diagram

The Activity Diagram represents the overall workflow of NutriLens, showing user actions and system responses from app launch to daily usage, including scanning products, logging meals, background risk analysis, and chatbot interactions.

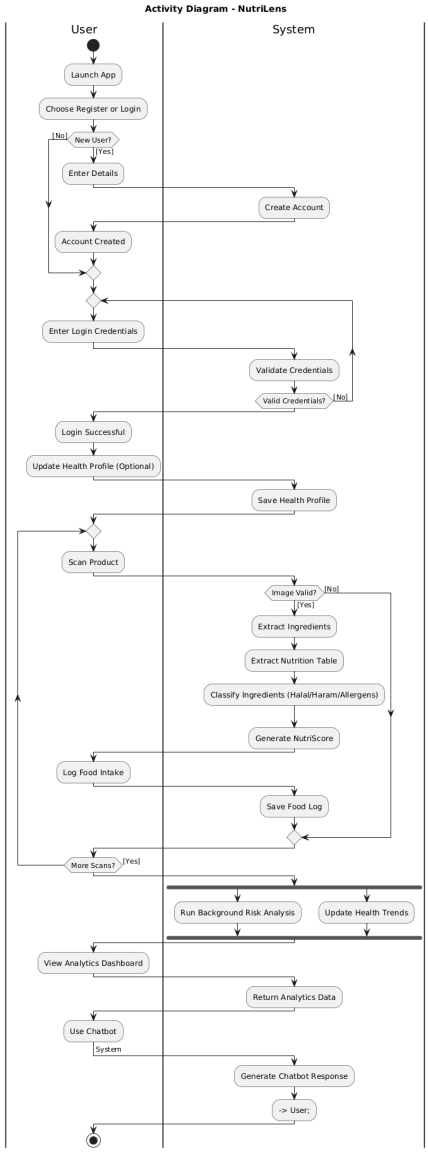

Figure 3.4: Activity Diagram of NutriLens

3.2.2 Class Diagram

The Class Diagram represents the static structure of NutriLens, illustrating the key system entities, their attributes, methods, and relationships. This helps visualize how modules such as User, Product, NutritionData, HealthProfile, and AI Nutrition Coach interact at a class level.

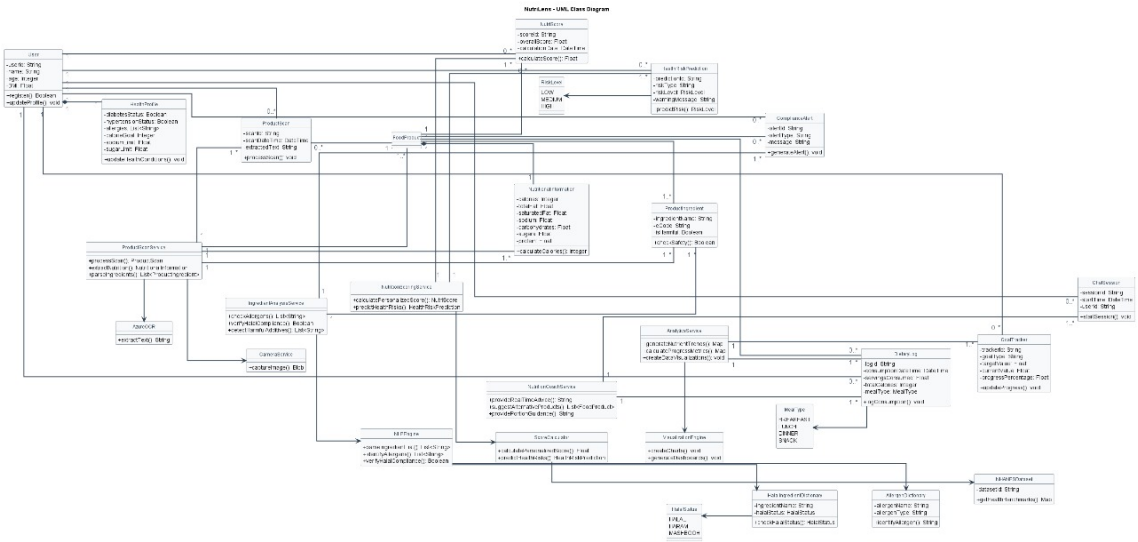

Figure 3.5: Class Diagram of NutriLens

3.2.3 System Sequence Diagram

This diagram shows high-level interactions between the user and the NutriLens system for key use cases such as registration, login, ingredient scanning, and dashboard access. It emphasizes the messages between the user and system.

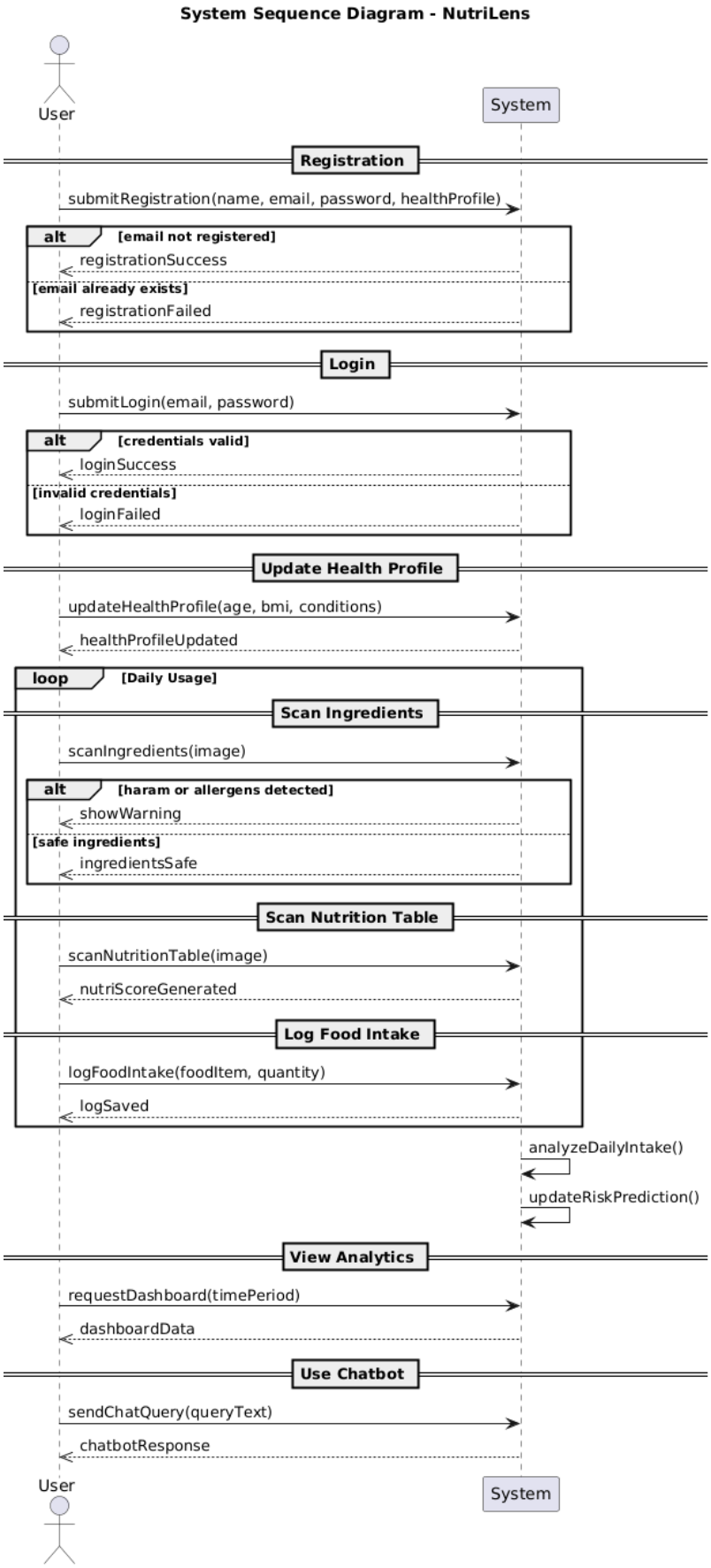

Figure 3.6: System Sequence Diagram of NutriLens

3.2.4 Action Sequence Diagrams

NutriLens has multiple Action Sequence Diagrams corresponding to each main use case. These diagrams detail the flow of messages between the user, UI, controller, business logic, database, and ML models.

3.2.4.1 User Registration

This diagram shows the user registration process, including validation of input data, checking if the email already exists, and storing new user details in the database.

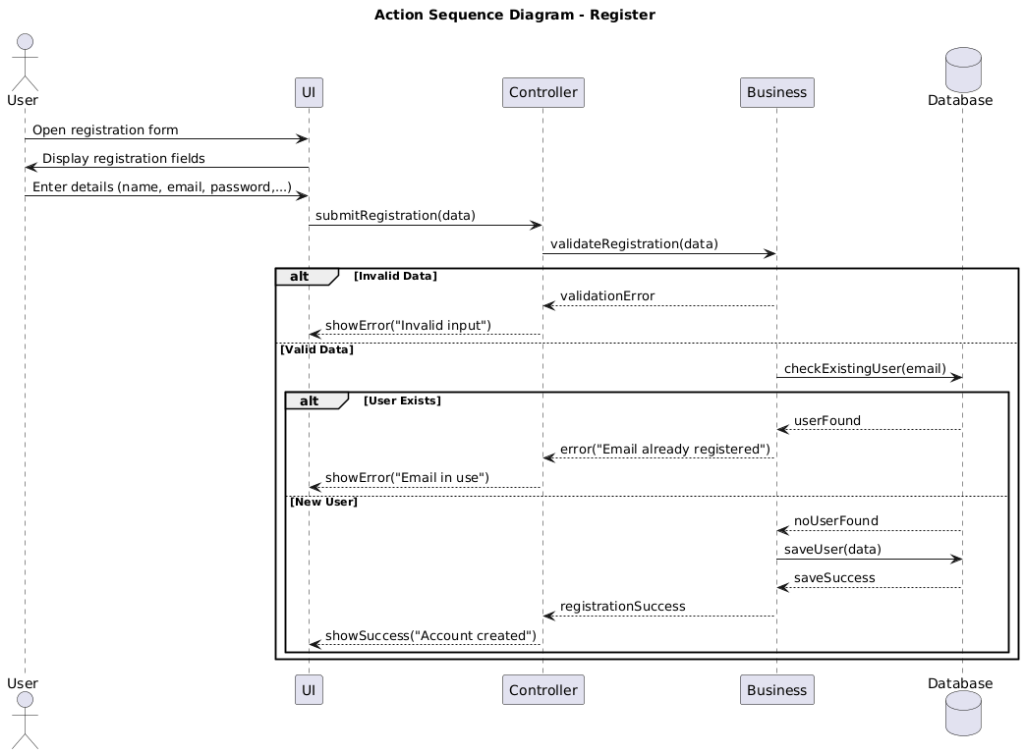

Figure 3.7: Action Sequence Diagram - User Registration

3.2.4.2 User Login

This diagram details user authentication, where the system validates credentials and generates a user session.

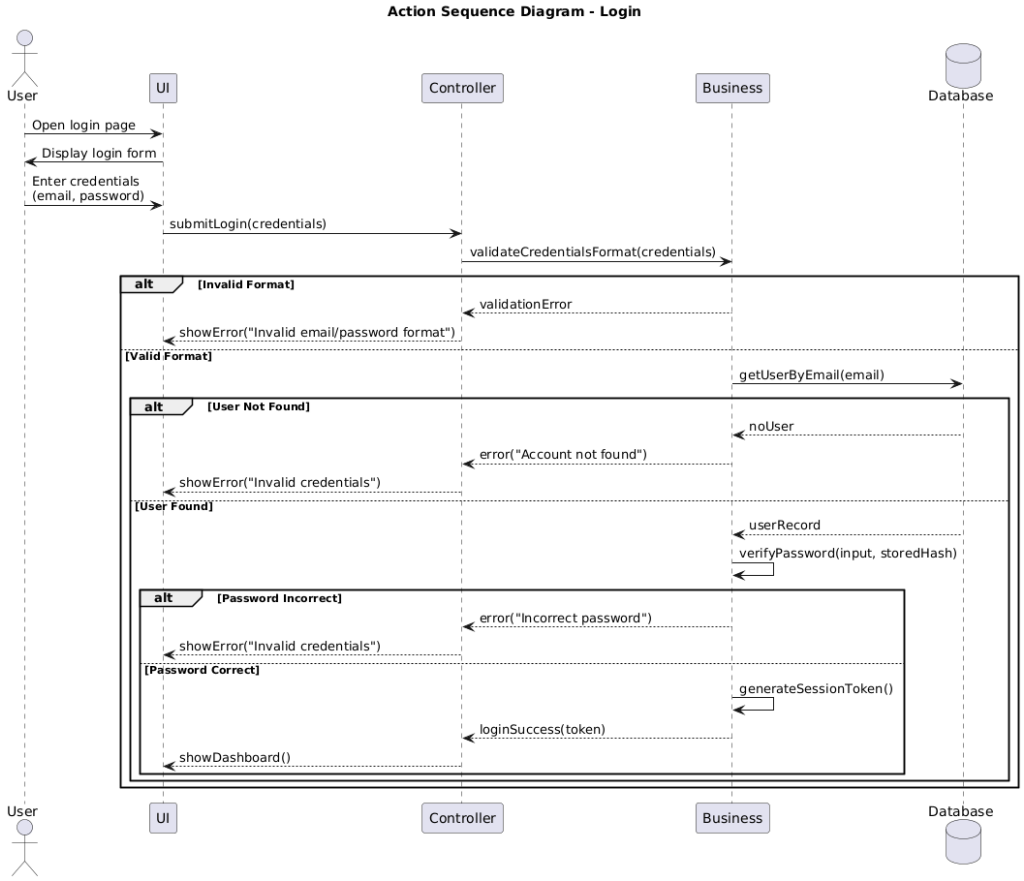

Figure 3.8: Action Sequence Diagram - User Login

3.2.4.3 Update Health Profile

This diagram shows the process of retrieving current health profile details and updating them in the database.

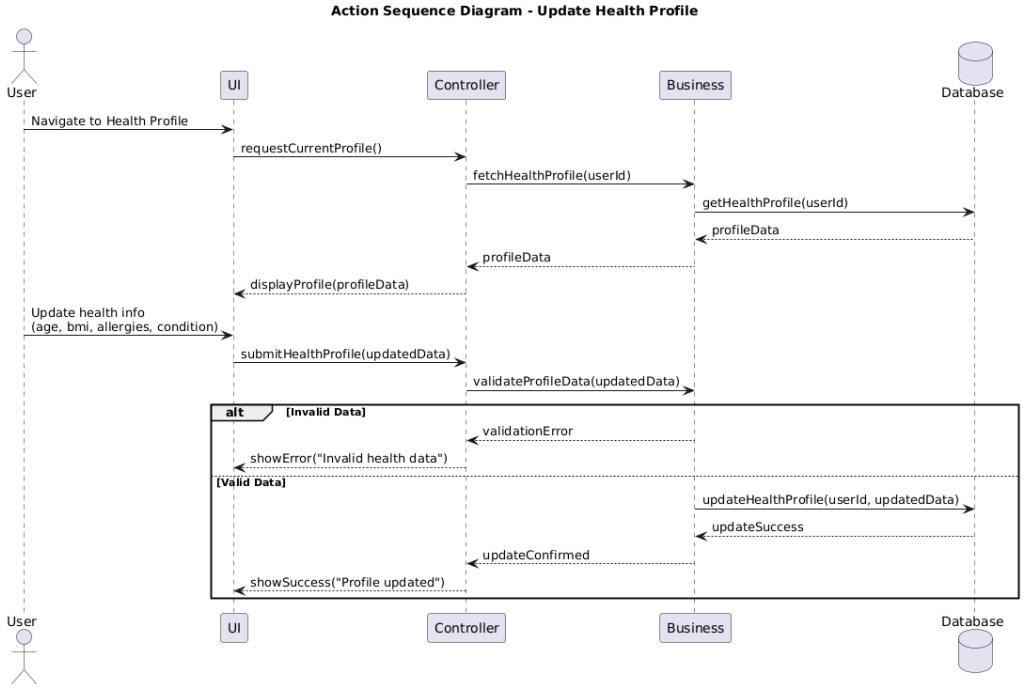

Figure 3.9: Action Sequence Diagram - Update Health Profile

3.2.4.4 Scan Ingredients

This diagram demonstrates OCR extraction of ingredients and classification of haram/allergen substances.

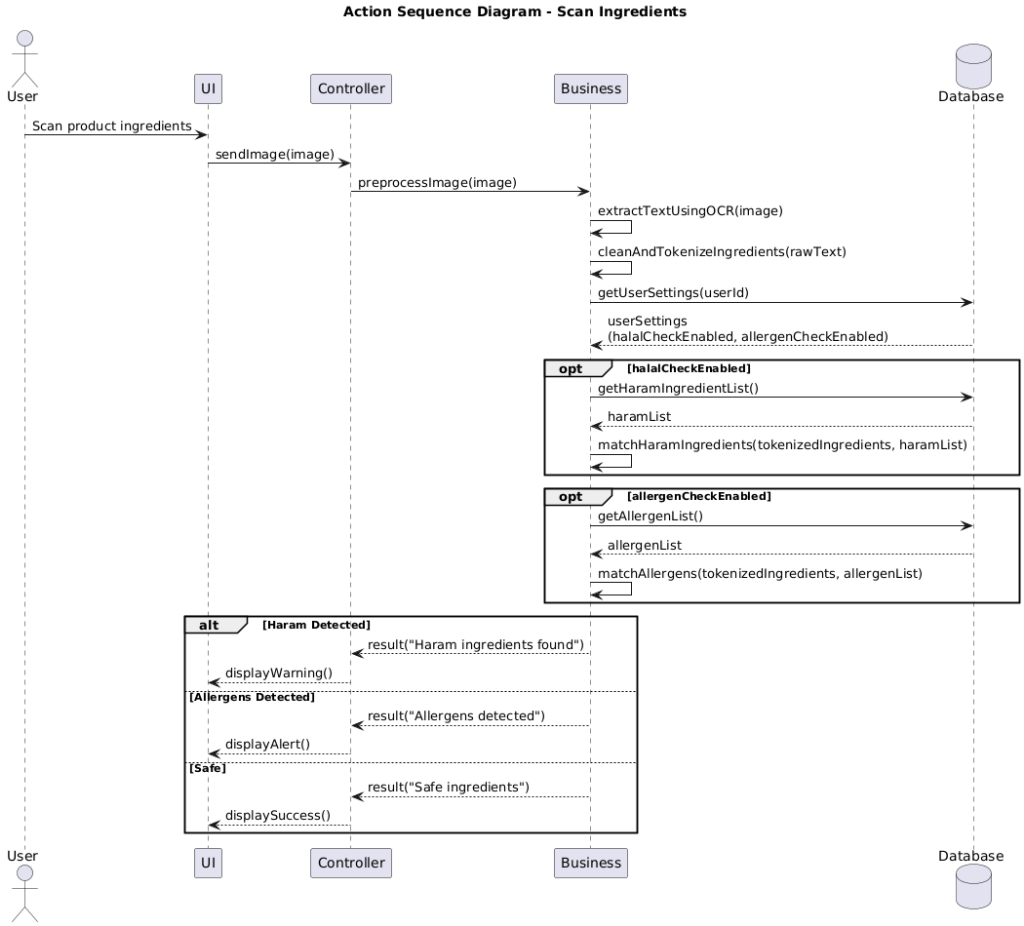

Figure 3.10: Action Sequence Diagram - Scan Ingredients

3.2.4.5 Generate Nutri-Score

This diagram shows OCR extraction of nutrition values and AI-based Nutri-Score generation.

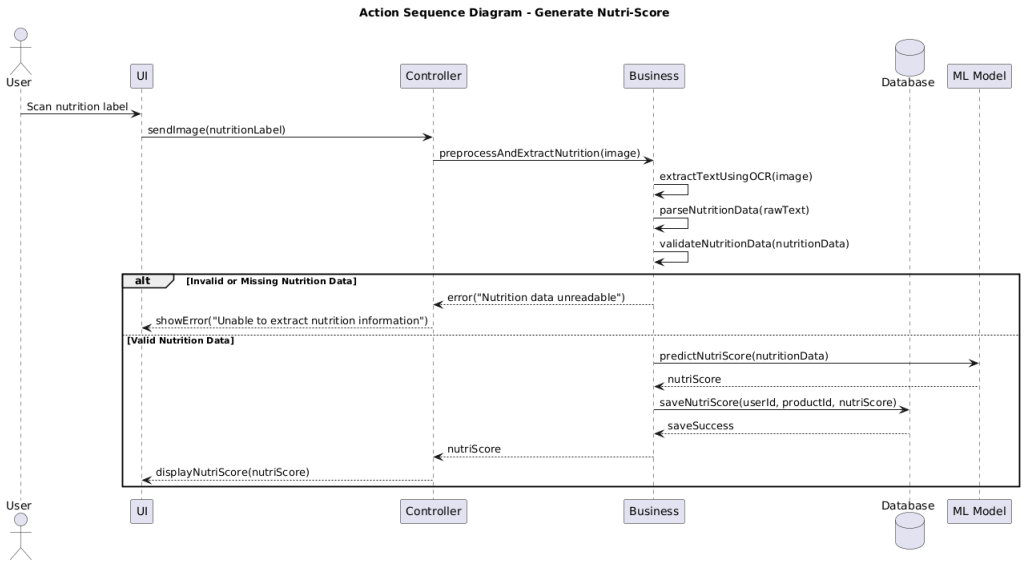

Figure 3.11: Action Sequence Diagram - Generate Nutri-Score

3.2.4.6 Log Food Intake

This diagram captures the process of logging food either from a previously scanned item or manual input.

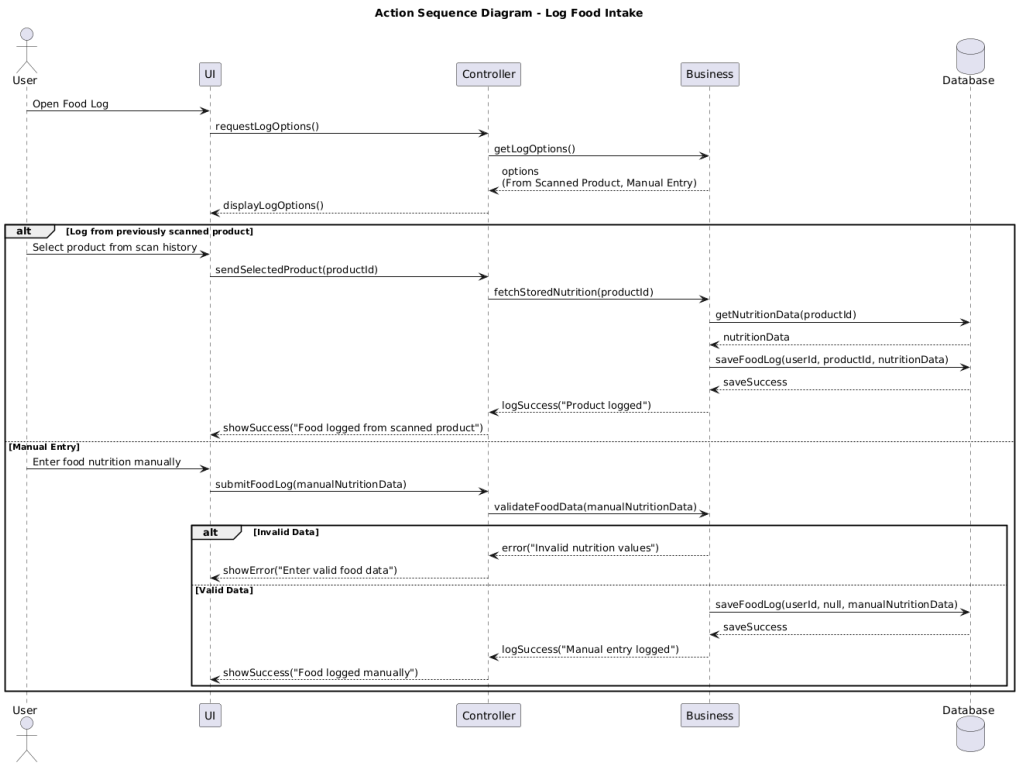

Figure 3.12: Action Sequence Diagram - Log Food Intake

3.2.4.7 Predict Health Risk

This diagram shows how daily nutritional logs are analyzed to predict chronic disease risks.

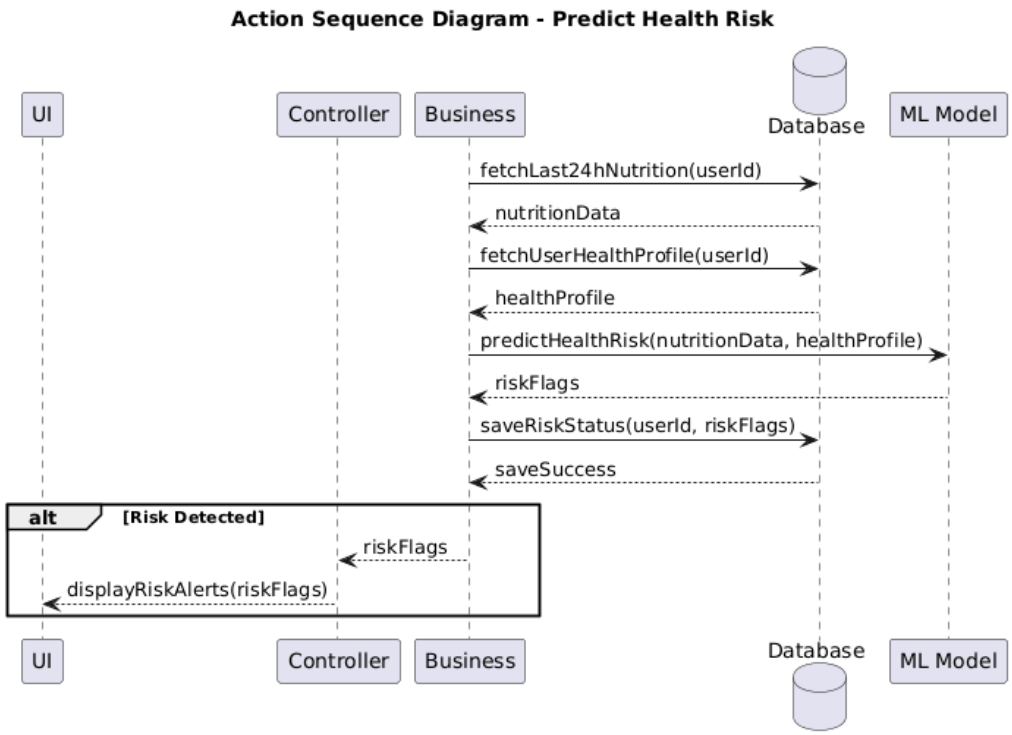

Figure 3.13: Action Sequence Diagram - Predict Health Risk

3.2.4.8 Analytics Dashboard

This diagram explains how dashboard visualizations are generated from user logs and risk data.

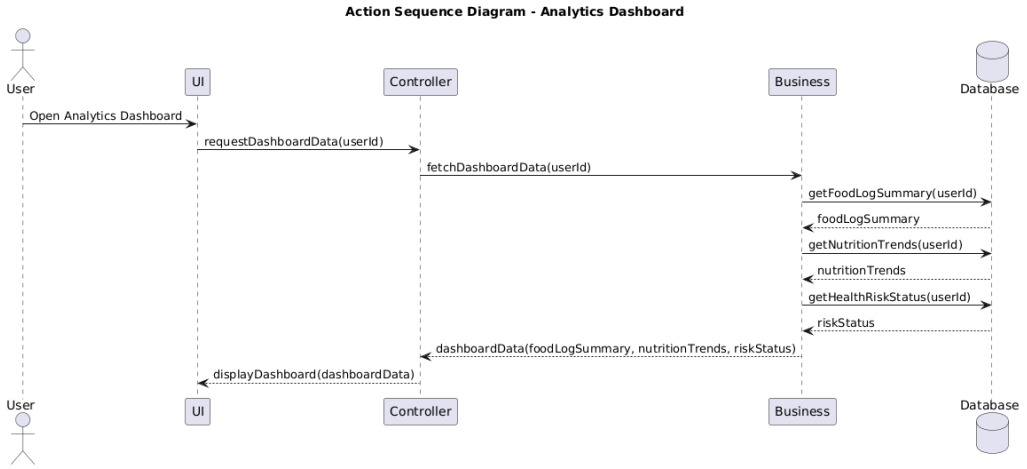

Figure 3.14: Action Sequence Diagram - Analytics Dashboard

3.2.4.9 Use AI Nutrition Coach

This diagram represents the interaction between the user and the AI chatbot for personalized nutrition guidance.

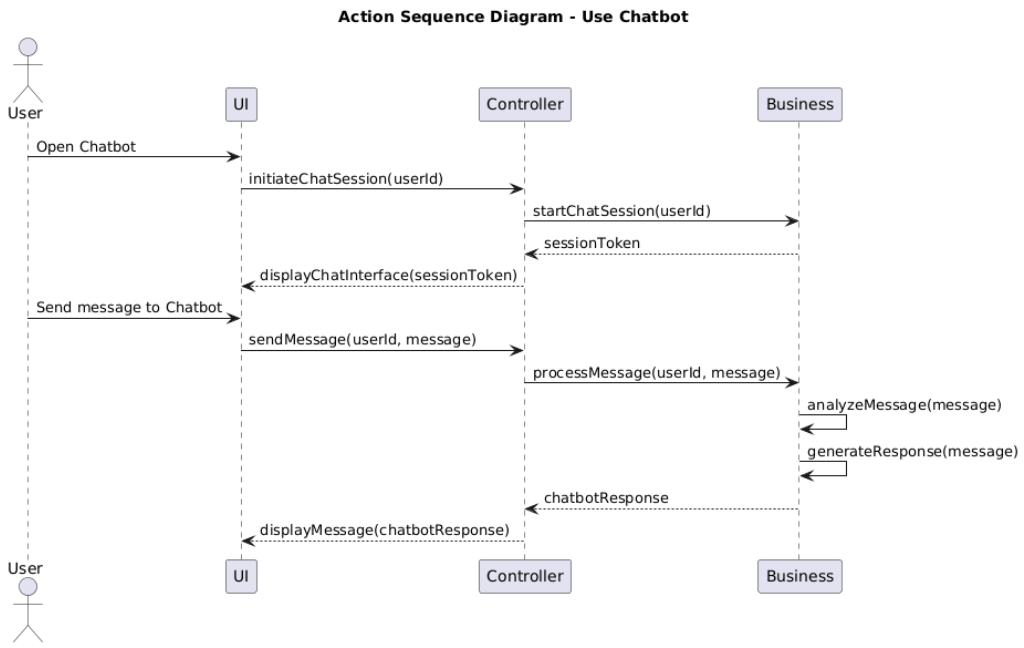

Figure 3.15: Action Sequence Diagram - Use AI Nutrition Coach

3.2.5 Data Flow Diagram

The Data Flow Diagram shows how data moves between the user, processes, and storage components. It illustrates all major system processes, external entities, and data stores, ensuring traceability of information.

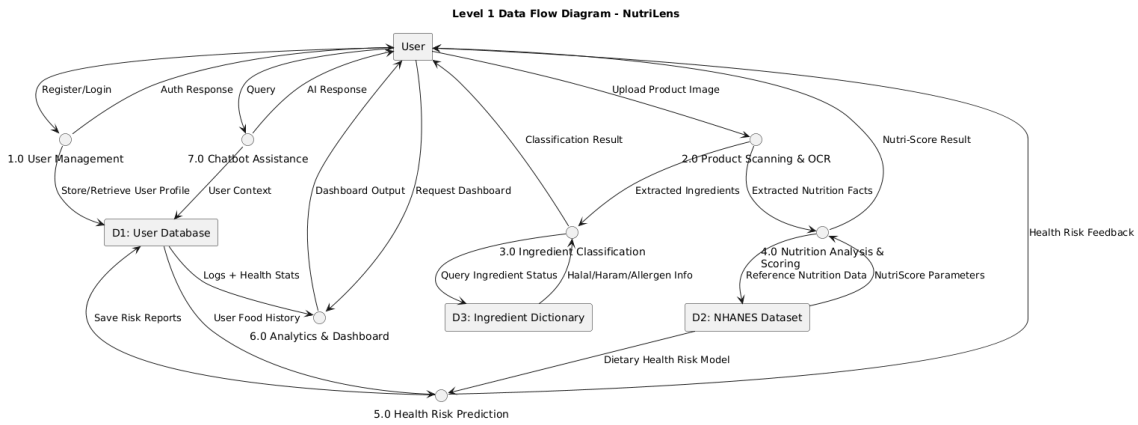

Figure 3.16: Level 1 Data Flow Diagram of NutriLens

3.2.6 State Transition Diagram

This diagram describes the state transitions of the product scanning process, including ingredient and nutrition scans, scan failures, and logging decisions. It highlights the dynamic behavior of the system in response to user actions.

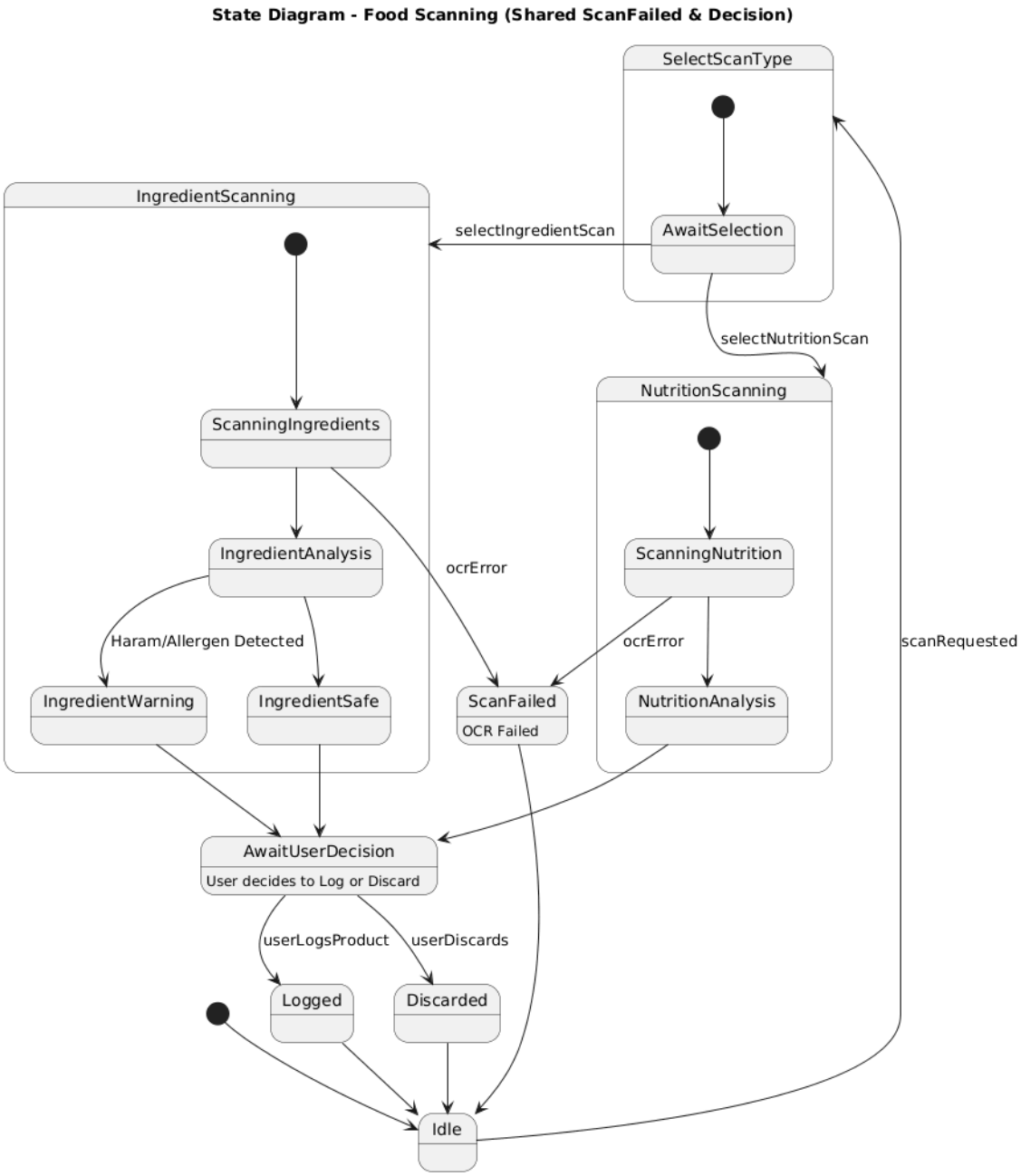

Figure 3.17: State Transition Diagram - Food Scanning in NutriLens

3.3 Data Design

The data design defines how information within the NutriLens system is structured, stored, and accessed to support its functional requirements. The system manages diverse data types such as user profiles, product information, ingredient classifications, nutritional values, machine learning outputs, and chatbot interaction logs. The aim is to ensure efficient retrieval, scalability, security, and integrity of data across all application modules.

3.3.1 Information Domain Transformation

The information domain of NutriLens consists of logical entities derived from key business operations such as user registration, ingredient scanning, meal logging, AI-based nutrition scoring, and health risk analysis. Each entity from the problem domain is transformed into structured data schemas using object-based modeling. The primary entities include:

- **User** – Represents personal, account, and authentication data.
- **Health Profile** – Stores user medical conditions, allergies, and dietary preferences.
- **Product** – Contains product details with nutritional values and extracted ingredients.
- **Food Log** – Maintains daily meal logs and calorie summaries.
- **Risk Prediction** – Stores machine learning generated health risk scores.
- **Chat History** – Logs user queries and responses from the AI coach.

Each entity is designed for loose coupling and high cohesion, promoting modularity and clean separation of concerns. Data normalization is applied where necessary to remove redundancy; however, controlled denormalization is selectively used to optimize read performance in frequently accessed collections such as user dashboard data.

3.3.2 Data Storage and Organization

NutriLens uses a modular schema design based on document-oriented data storage, which allows flexible storage of hierarchical user-related data. This data model aligns well with JSON-like structures used in modern mobile applications and REST APIs.

Entity Name	Description
Users	Stores credentials, roles, and profile information
HealthProfiles	Contains health metrics, allergies, and BMI calculations
Products	Stores product metadata, ingredients and nutrition data
ScannedResults	OCR extracted text and ingredient classification
FoodLogs	Daily food intake entries linked to users
RiskScores	Machine learning generated health risk predictions
ChatSessions	Logs AI chatbot conversations

Table 3.1: Core Data Collections in NutriLens

Each collection maintains a unique identifier to support traceability and integrity. Referential links are managed using document references (e.g. `user_id`, `product_id`) to establish logical relationships between entities.

3.3.3 Entity Relationship Diagram

The entity structure and relationships of the NutriLens database are represented in Figure 3.18.

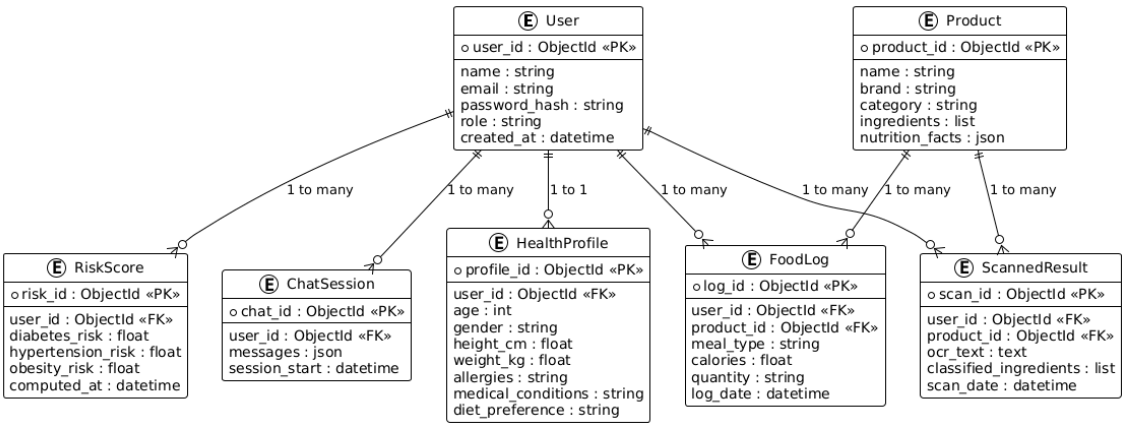

Figure 3.18: Entity Relationship Diagram (ERD) of NutriLens

3.3.4 Data Processing

Data flows through three primary layers in NutriLens:

- **Input Layer** – Raw inputs such as user registration data and scanned images.

- **Processing Layer** – OCR preprocessing, NLP ingredient classification, and ML-based risk evaluation.
- **Storage Layer** – Validated and transformed data stored in respective collections.

Example workflow:

> User scans a product → OCR extracts text → NLP classifies ingredients → Nutri-Score computed → log saved in database.

3.3.5 Database and Storage Technology

A NoSQL database (MongoDB) is used due to:

- Schema flexibility for evolving requirements
- High scalability and read performance
- Natural fit with JSON-based APIs
- Horizontal scaling with sharding and replication

3.3.6 Summary

The data design of NutriLens follows a modular and scalable structure optimized for user analytics, AI interactions, and real-time product scanning. The document-based schema supports rapid development while ensuring consistency across related entities.

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